Ignacio Dassori

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PROFILE

Robotics and AI engineer specializing in computer vision, deep learning, and autonomous systems. Experienced in developing robotic software, perception pipelines, and machine learning solutions for industrial and research applications. Currently finishing an M.Sc. in Robotics, Cognition, and Intelligence at the Technical University of Munich.

EXPERIENCE

Maquintel SpA — Apr. 2024 – Sep. 2024
Robotics Software Engineer — Santiago, Chile

- Developed robotics software for autonomous underground inspection platforms deployed in mining environments, focusing on perception, data processing, and system reliability.
- Implemented multi-modal perception pipelines combining LiDAR and RGB data for automated anomaly detection and infrastructure assessment.
- Generated automated inspection reports and reconstructed 3D assets from sensor data to support client decision-making and maintenance planning.
- Developed GPS-guided unmanned ground vehicle (UGV), including navigation and control software, enabling autonomous operation in outdoor mining environments.

PSINet — Jan. 2023 – Dec. 2023
Computer Vision Engineer — Santiago, Chile

- Started as an intern and transitioned into a part-time engineering role, contributing to production computer vision systems for industrial safety applications.
- Built and curated large-scale computer vision datasets for training automated risk detection models in mining environments.
- Developed a dashboard for visualization and validation of detected incidents.
- Designed and implemented an end-to-end computer vision pipeline covering real-time video ingestion, model inference, and database integration. .
- Optimized and deployed YOLOX-based object detection models through CCTV camera feed for real-time inference.

Technical University of Munich — Apr. 2025 – Jul. 2026
Introduction to Deep Learning Tutor — Munich, Germany

- Supported the delivery of TUM's *Introduction to Deep Learning* course, mentoring 1,000+ students through weekly office hours, technical discussions, and programming support.
- Designed and maintained practical deep learning assignments, evaluated student submissions, and contributed to the development and grading of the final examination.

Advanced Mining Technology Center — Jan. 2021 – May. 2021
Internship — Santiago, Chile

- Developed realistic simulated underground environments in Gazebo for evaluating robotics algorithms.
- Evaluated and benchmarked SLAM algorithms for autonomous exploration, analyzing performance across simulated underground scenarios.

EDUCATION

Technical University of Munich — Oct. 2024 – Expected Sept. 2026
M.Sc. Robotics, Cognition, Intelligence — Munich, Germany

- GPA: 1.5/4.0 (German scale: 1.0 best - 4.0 worst)

Universidad de Chile — Mar. 2018 – Feb. 2024
Bachelor of Science in Engineering Sciences, Major in Electrical Engineering — Santiago, Chile

- GPA: 6.6/7.0 (Chilean scale: 7.0 best - 1.0 worst)

PROJECTS

Kinect Fusion Re-Implementation Dec. 2025 – Feb. 2026

- Semester project for course *3D Scanning and Motion Capture* at TUM.
- Reimplemented the Kinect Fusion pipeline for real-time RGB-D reconstruction, leveraging GPU acceleration and CUDA optimization.
- Frame-to-model camera pose tracking using point-to-plane iterative closest point.
- Fusing of RGB-D sensor data into a dense volumetric model (TSDF).
- Developed custom CUDA kernels achieving real-time reconstruction performance at 47 FPS.

FoamSurf: Novel View Synthesis Sept. 2025 – Nov. 2025

- Semester project for course *Advanced Deep Learning for Computer Vision* at TUM.
- Lightweight extension of the *Radiant Foam (RadFoam)* paper. Addition of monocular geometric priors to improve surface reconstruction quality.
- Achieved improvement in mesh quality metrics over base RadFoam and other SoTA methods.

RoboTUM Senior Member Oct. 2024 – Present

- Core member of RoboTUM, leading interdisciplinary robotics projects involving 40+ students across software, hardware, and AI domains.
- Lead hardware and software integration of an autonomous navigation system for a biped robot, including Jetson onboard compute, LiDAR sensor fusion, and real-time mapping and localization.
- Secured €10,000 in external sponsorship funding through industry partnerships.
- Organized technical workshops covering ROS2, Git, reinforcement learning, and embedded robotics development.

Research Publication: Quadruped Robot Locomotion Sept. 2023 – May 2024

- Published paper titled *Four-Legged Gait Control via the Fusion of Computer Vision and Reinforcement Learning* at the 27th International Conference on Information Fusion 2024.
- Developed a reinforcement learning framework using Proximal Policy Optimization (PPO) to adapt quadruped robot locomotion policies.
- Applied autoencoder-based representation learning to extract compact visual features from RGB observations for robot control.

SKILLS

Languages	Spanish (native), English C1, German C1
Programming	Python, C++, CMake, CUDA, C, Matlab
Machine Learning	PyTorch, TensorFlow, OpenCV, NumPy, Computer Vision, Deep Learning, Reinforcement Learning, 3D Reconstruction, GTSAM
Robotics	ROS/ROS2, SLAM, Sensor Fusion, Gazebo, MuJoCo, PyBullet, Motion Planning, Robot Control
Hardware	Jetson Orin Nano, Jetson Thor, Raspberry Pi, LiDAR, RGB-D Cameras, IMU, GNSS, UR10, Clearpath Husky, Soldering, Cabling Assembly, 3D Printing, Quick Prototyping

HONORS AND AWARDS

Scholarship Technical University Munich

- Awarded merit-based waver scholarship for international students.

Universidad de Chile

- Outstanding Student of the Faculty of Physical and Mathematical Sciences [2018, 2019, 2020, 2021, 2022, 2023].